Journal of Pediatric Surgery (2010) **45**, 693–697

ELSEVIER

Journal of Pediatric Surgery

www.elsevier.com/locate/jpedsurg

Early repair of congenital diaphragmatic hernia on extracorporeal membrane oxygenation

Melvin S. Dassinger[a,*], Daniel R. Copeland[a], Jeff Gossett[b], Danny C. Little[a], Richard J. Jackson[a], Samuel D Smith[a]
The Congenital Diaphragmatic Hernia Study Group[c]

[a]*Pediatric Surgery, University of Arkansas for Medical Sciences, Little Rock, AR, USA*
[b]*Biostatistics Program, Department of Pediatrics, Little Rock, AR, USA*
[c]*Pediatric Surgery, University of Texas Health Sciences Center, Houston, TX, USA*

Received 24 May 2009; revised 6 August 2009; accepted 7 August 2009

Key words:
Congenital diaphragmatic hernia;
Extracorporeal membrane oxygenation

Abstract
Background: Timing of repair of congenital diaphragmatic hernia (CDH) in babies that require stabilization on extracorporeal membrane oxygenation (ECMO) remains controversial. Although many centers delay operation until physiologic stabilization has occurred or ECMO is no longer needed, we repair soon after ECMO has been initiated. The purpose of this study is to determine if our approach has achieved acceptable morbidity and mortality.
Methods: Charts of live-born babies with CDH treated at our institution between 1993 and 2007 were retrospectively reviewed. Data were then compared with The Congenital Diaphragmatic Hernia Study Group and Extracorporeal Life Support Organization registries.
Results: Forty-eight (39%) patients required ECMO Thirty-four of these 48 neonates were cannulated before operative repair. Venoarterial ECMO was used exclusively. The mean (SD) time of repair from cannulation was 55 (21) hours. Survival for this subset of patients was 71%. Three patients (8.8%) who underwent repair on ECMO experienced surgical site hemorrhage that required intervention.
Conclusion: Early repair of CDH in neonates on ECMO can be accomplished with acceptable rates of morbidity and mortality.

Congenital diaphragmatic hernia (CDH) is a complex anomaly affecting 1 in 2 to 4000 live births. The hallmarks of this disease are severe respiratory failure secondary to pulmonary hypoplasia and persistent pulmonary hypertension [1]. Treatment strategies have evolved from the use of paralytic agents along with induced alkalosis to more modern approaches, which attempt to minimize barotrauma via permissive hypercapnia in a spontaneously breathing patient. Centers that used these techniques saw both a decreased pneumothorax rate and an increased overall survival [2-4]. Furthermore, these strategies, coupled with the use of sophisticated ventilators, have led to a decreased need for extracorporeal support [5,6].

* Corresponding author. Division of Pediatric Surgery, Arkansas Children's Hospital, Little Rock, AR 72202, USA. Tel.: +1 501 3641447; fax: +1 501364 1516.
E-mail address: dassingermelvins@uams.edu (M.S. Dassinger).

0022-3468/$ – see front matter © 2010 Elsevier Inc. All rights reserved.
doi:10.1016/j.jpedsurg.2009.08.011

Despite these advances, many neonates with CDH and persistent pulmonary hypertension require further support to provide adequate tissue oxygenation and perfusion. Extracorporeal membrane oxygenation (ECMO) is reserved for these patients, with cannulation often necessary before correction of the defect. In these cases, timing of repair, or even if repair should be offered, while on ECMO remains controversial. The fear of hemorrhagic complications in a systemically anticoagulated patient has led many centers to plan operative intervention when ECMO therapy is nearing completion; others do not offer operation unless ECMO is no longer needed, leading to a small percentage of babies who are never repaired. Conversely, our institution has traditionally attempted repair of the diaphragmatic hernia within 72 hours of commencement of extracorporeal support. The purpose of the current study is to determine if "early" repair of CDH on ECMO can be accomplished with acceptable morbidity and mortality.

1. Materials and Methods

1.1. Design

Approval for this study was obtained through the Institutional Review Board for the University of Arkansas for Medical Sciences (#100342). Charts of newborns diagnosed with CDH at Arkansas Children's Hospital between January 1993 and December 2007 were reviewed. Data were recorded in accordance with guidelines for enrollment in The Congenital Diaphragmatic Hernia Study Group (CDHSG) and included demographics, birth history, associated anomalies, pharmacologic manipulation, ventilator management, operative repair, and postoperative complications. Particular attention was paid to those children who required ECMO before operative repair.

1.2. Statistical analysis

Overall survival, timing of repair, and length of cannulation data were then compared with the CDHSG registry. Bleeding complication rates were compared with babies with CDH repaired on ECMO between 1993 and 2007 listed in the Extracorporeal Life Support Organization (ELSO) database. Fisher's Exact test was used to analyze categorical variables, whereas the nonparametric Wilcoxon rank sum test was used for nonnormally distributed data. Significance was based on a P value of .05. All calculations were performed using Stata version 10.1 (Stata, College Station, Tex).

Fourteen neonates with CDH and significant comorbidities underwent neither repair of the diaphragmatic defect nor ECMO cannulation. These patients were excluded from statistical analysis.

1.3. Protocol

Repair of the diaphragmatic defect is typically performed within 72 hours of cannulation. Activated clotting time (ACT) is kept between 140 and 160 seconds in the operative and immediate perioperative period. After 24 to 48 hours, if there is no evidence of bleeding, the ACT is maintained at 180 to 200 seconds. In addition, aminocaproic acid (Amicar) is administered before incision in the form of a bolus of 100 mg/kg followed by an infusion of 33 mg kg^{-1} h^{-1} for 72 postoperative hours. Platelets are kept more than 100,000 for the duration of ECMO therapy.

2. Results

Forty-eight (39%) patients were placed on ECMO, with venoarterial ECMO used exclusively. Thirty-four of 48 required ECMO preoperatively including 2 patients who were cannulated by our mobile ECMO team at other institutions and transported by air to our facility. Thirty-four (100%) of 34 patients subsequently underwent repair of the diaphragmatic hernia. Forty-one percent of babies were diagnosed prenatally, with the majority of these delivered at the university obstetrics unit affiliated with our children's hospital. Twenty-five (74%) of 34 were delivered vaginally. Mean gestational age of this cohort was 37 weeks, with an average birth weight of approximately 3 kg. Roughly one quarter were found to have nonlethal cardiac anomalies on initial echocardiogram.

Mean length of operative repair was 115 minutes (range, 39-240 minutes). Defects were noted to be left sided in 29, right sided in 3, and centrally located/bilateral in 2. Seventeen (50%) of 34 neonates were found to have liver displaced into the chest, and 79% required a prosthetic closure of the defect. All babies were approached via a subcostal incision. Four patients required a temporary patch closure of the abdominal wall, whereas a temporary silo was placed in 1 patient.

Table 1 Early Repair of CDH on ECMO

	No. repaired	Days after cannulation	Days on ECMO	Survival	Surgical site bleed
Our institution	34/34 (100%)	2.3 (±1.94)	11.7 (±2.4)	24/34 (71%)	3/34 (8.8%)
CDHSG/ELSO *	3607/4173 (86%)	6.9 (±0.43)	13.3 (±0.56)	317/653 (49%)	905/5727 (14.7%)*
P	<.0001	<.0001	.15	.016	.35

* Data from the ELSO registry.

Neonates at our institution were repaired earlier in the course of the ECMO run compared with the CDHSG registry (2.4 versus 7 days, $P < .0001$; Table 1). The median day of repair after cannulation for our institution and the CDHSG was 1 and 6, respectively (Fig. 1). There was no statistical difference in length of time required on ECMO between the 2 groups (11.7 versus 13.3 days, $P = .15$). The median length of cannulation was 10 days at Arkansas Children's Hospital compared with 12 days in the CDHSG.

Twenty-four (71%) of 34 patients in this subset of patients survived compared with the CDHSG database rate of 50.9% ($P = .016$). In all 10 patients who did not survive, the surgeons' notes describe the defect as "large," "very large," or "near agenesis." One of these cases was complicated by iatrogenic esophageal perforation from nasogastric tube placement in the immediate postoperative period.

Eleven (32%) of the 34 patients experienced a bleeding complication during their ECMO therapy (Table 2). Six of these patients experienced intracranial hemorrhage, whereas 2 experienced subgaleal bleeds. Only 3 patients (8.8%) experienced a surgical site hemorrhage after repair on ECMO, which is comparable to the ELSO registry (14.7%, $P = .35$). Four bleeds necessitated intervention including 2 tube thoracostomies, 1 evacuation of a cephalohematoma, and 1 small bowel resection. No deaths were directly attributable to hemorrhage.

Twenty of 34 patients received aminocaproic acid (Amicar) infusions, including 6 of the 11 who experienced bleeding complications. No patients experienced thrombotic complications. Extracorporeal membrane oxygenation circuits were changed in 13 patients (38%), with a mean of 8.85 days after cannulation (range, 4-14 days; median, 9 days).

Twenty-three (68%) of 34 underwent a second operation during their initial hospital stay, including 14 fundoplications for reflux. There has been only one reoperation for recurrence documented in this cohort.

Table 2 Bleeding complications on ECMO

Patient	Location of hemorrhage	Outcome
1	Cerebellar	Resolved
2	Subdural	Resolved
3	Grade I IVH	Resolved
4	Cephalohematoma	Evacuation of hematoma
5	Small bowel hematoma	Small bowel resection
6	Grade 4 IVH	Cystic encephalomalacia
7	Abdomen	No intervention
8	Left chest	Chest tube
9	Left chest	Chest tube
10	Abdomen	No intervention
11	Grade I IVH	Resolved

IVH indicates intraventricular hemorrhage.

The 14 patients who were not cannulated for ECMO and had no subsequent correction of the diaphragmatic defect had comorbidities including complex congenital heart disease, chromosomal abnormalities, and cardiopulmonary collapse (Table 3).

In the past 15 years, 124 neonates with CDH underwent operative repair at our institution with an overall survival of 87%.

3. Discussion

The approach to operative management of neonates with CDH has evolved since originally described by Ladd and Gross [7]. Initially felt to be a surgical emergency, delayed repair has now become widely accepted, with timing of

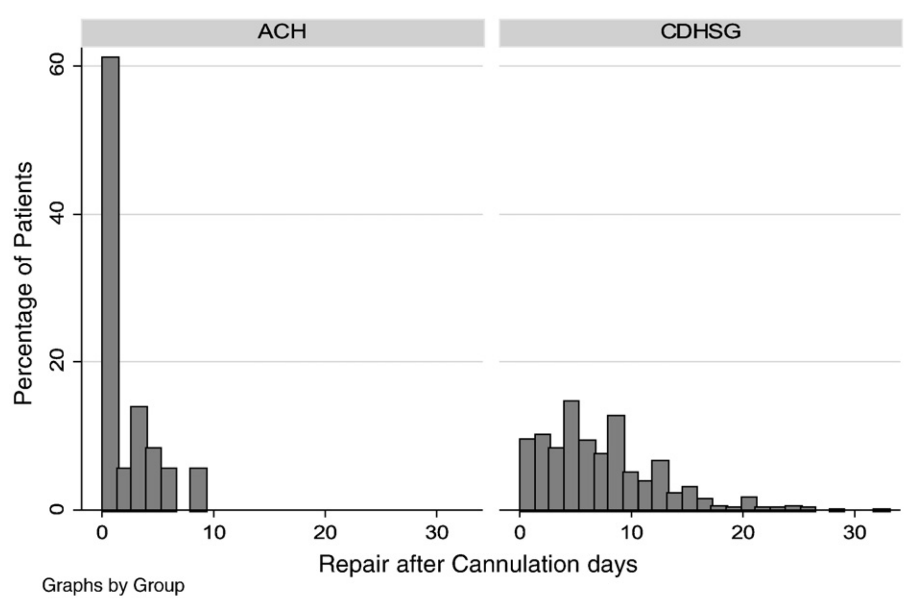

Fig. 1 Median day of repair after cannulation was day 1 at Arkansas Children's Hospital compared with day 6 for the CDHSG.

Table 3 Neonates with significant comorbidities offered neither ECMO nor repair of diaphragmatic hernia

Patient	Comorbidity
1	Multiple congenital anomalies
2	Trisomy 22, multiple anomalies
3	Multiple congenital anomalies, chromosomal translocations
4	Hypoplastic left heart syndrome
5	Trisomy 13, double-outlet right ventricle
6	Diffuse encephalopathy after 90 minutes of cardiopulmonary arrest/resuscitation
7	Hypoplastic left heart syndrome
8	Hemodynamic instability owing to bilateral pneumothoraces
9	Trisomy 18, coarctation with hypoplastic aortic arch
10	Hypoplastic left heart syndrome
11	Fryns syndrome, double-outlet right ventricle
12	Arrested during transport secondary to tension pneumothorax
13	Arrested after bilateral pneumothoraces at outside facility
14	Hypoplastic left heart syndrome

operation based on physiologic stabilization [8]. For those babies unable to adequately oxygenate or ventilate because of persistent fetal circulation, ECMO can be a life-saving measure; multiple centers have reported that ECMO contributed to survival in these babies with CDH who would have otherwise not been salvageable [9,10]. However, using ECMO in these high-risk neonates presents unique management challenges.

The first fundamental question that must be answered is whether ECMO should be offered. Patients with concomitant severe cardiac anomalies, especially single ventricle physiology, have a much poorer prognosis than those babies with isolated diaphragmatic defects [11]. Similarly, neonates with diaphragmatic hernia associated with a syndrome, for example, Fryns syndrome, or multiple congenital anomalies also have a grim prognosis [12]. A multidisciplinary approach, including neonatologists, geneticists, cardiologists, and surgeons, is critical for the depth of parental counseling necessary. The 14 patients not cannulated for ECMO in our series were felt to be nonsurvivors, and the families elected to have no intervention performed.

Once placed on ECMO, controversy regarding the optimal time to perform the operation exists. Neonates at our institution underwent operative repair an average of 2.4 days (55 hours) into the ECMO course, 5 days earlier than those reported in the CDHSG. No operations were delayed until ECMO decannulation. In comparison, 27% of patients registered with the CDHSG who required ECMO had operative repair after the run was completed.

We feel that early repair offers 2 potential advantages. First, babies tend to develop anasarca as the number of days on ECMO proceeds. Performing an operation before the development of body wall edema allows for a technically easier operation. Although not a direct measurement of the degree of edema, the ability to primarily close the abdomen is limited after anasarca has developed. Twenty-nine (85%) of our 34 patients were able to have primary fascial closure, whereas 32% of babies repaired on ECMO at other facilities required nonprimary abdominal wall closures. Second, recovery from the physiologic insult of an operation earlier in the ECMO course might lead to fewer total days on ECMO. There was a trend toward fewer days required on ECMO noted at our institution, although this did not reach statistical significance.

Hesitancy to perform repair while systemically anticoagulated on ECMO undoubtedly stems from an increased rate of hemorrhage, which has been associated with increased mortality. This was not seen in the current study. Bleeding complications were seen in 32% (11/34) of our patients repaired on ECMO, comparing favorably to previously published reports [13]. Moreover, our surgical site hemorrhage rate was comparable to that recorded in the ELSO registry. Only 4 of these required surgical interventions to control the hemorrhage and none of the bleeding episodes were directly attributable to any deaths. Our low hemorrhage rates are certainly related to our perioperative management protocol and prompt correction of coagulopathies. Similar low rates of surgical site hemorrhage have been noted at other institutions [14].

The risk of thrombotic complications, both in the patient and in the circuit, is theoretically increased with use of an inhibitor of fibrinolysis in conjunction with low ACTs. However, neither cerebral infarction nor large vessel thrombus was documented in our patient population. Similar findings were reported by both Downard et al and Horwitz et al [14,15]. In addition, only 13 of 34 patients had circuit changes during their ECMO course, with an average time to circuit change of 8 days. This duration is anecdotally similar to circuits used for ECMO in neonates who do not require operative intervention at our institution and consequently do not require lower ACTs.

Not all babies will undergo operative correction of their diaphragmatic defects. Of the 1424 babies who required ECMO in the CDHSG registry, 219 (15.3%) never underwent operative repair. This number likely includes babies treated at centers where operative repair is not undertaken unless the baby can be weaned off extracorporeal support. In contrast, all 34 babies at our institution who were placed on ECMO were subsequently repaired.

It is difficult for even the most experienced clinician to predict outcomes either before cannulation or during ECMO therapy. This fact has led to attempts to predict survival based on physiologic parameters and laboratory values before any intervention. Risk stratification of babies with CDH has been proposed based on apgar scores and birth weight [16]; however, prospective studies applying this stratification have not been performed. Furthermore, Heiss and Clark [17] attempted unsuccessfully to determine variables that would predict greater than 90% mortality in neonates with CDH despite use of ECMO [17]. At present, there are no uniformly applicable criteria that allow practitioners to effectively determine nonsurvivors, arguing for liberal criteria for repair.

Many centers rely on a single surgeon or physician directing all patient care for babies with CDH. Conversely, our multidisciplinary approach involves multiple surgeons, intensivists, and practitioners. Although this approach has led to low morbidity and mortality, specific reasons for these good outcomes, which were achieved using differing treatment strategies, are difficult to ascertain via a retrospective review. In addition, criteria for placement on ECMO varied over the years, although our rate of use is similar to that reported by the CDHSG. Furthermore, we recognize that comparisons with large databases are limited by each centers' interpretation of complications, for example, surgical site hemorrhage, and practice patterns. However, databases reflect international treatment patterns, and comparisons to these are more meaningful than single institution data reporting, especially given our relatively small numbers. Lastly, we recognize that timing of repair is only part of the management of these challenging patients. Despite these limitations, early repair on ECMO has yielded acceptable mortality and morbidity rates and may be considered in neonates with CDH requiring extracorporeal support.

References

[1] Geggel RL, Murphy JD, Langleben D, et al. Congenital diaphragmatic hernia: arterial structural changes and persistent pulmonary hypertension after surgical repair. J Pediatr 1985;107:457-564.
[2] Wung JT, Sahni R, Moffitt ST, et al. Congenital diaphragmatic hernia: survival treated with very delayed surgery, spontaneous respiration, and no chest tube. J Pediatr Surg 1995;30(3):406-9.
[3] Boloker JD, Bateman DA, Wung JT, et al. Congenital diaphragmatic hernia in 120 infants treated consecutively with permissive hypercapnia/spontaneous respiration/elective repair. J Pediatr Surg 2002;37(3):357-66.
[4] Kays DW, Langham MR, Ledbetter DJ, et al. Detrimental effect of standard medical therapy in congenital diaphragmatic hernia. Ann Surg 1999;230(3):340-51.
[5] Datin-Dorrier V, Walter-Nicolet E, Rousseau V, et al. Experience in the management of eighty-two newborns with congenital diaphragmatic hernia treated with high-frequency oscillatory ventilation and delayed surgery without the use of extracorporeal membrane oxygenation. J Intensive Care Med 2008;23(2):128-35.
[6] Migliazza L, Bellan C, Alberti D, et al. Retrospective study of 111 cases of congenital diaphragmatic hernia treated with early high-frequency oscillatory ventilation and presurgical stabilization. J Pediatr Surg 2007;42:1526-32.
[7] Ladd WE, Gross RE. Congenital diaphragmatic hernia. N Eng J Med 1940;223:917-25.
[8] Reyes C, Chang LK, Waffarn F, et al. Delayed repair of congenital diaphragmatic hernia with early high-frequency oscillatory ventilation during preoperative stabilization. J Pediatr Surg 1998;33(7):1010-6.
[9] Heiss K, Manning P, Oldham K, et al. Reversal of morbidity for congenital diaphragmatic hernia with ECMO. Ann Surg 1989;209:225-30.
[10] West KW, Bengston K, Rescorla FJ, et al. Delayed surgical repair and ECMO improves survival in congenital diaphragmatic hernia. Ann Surg 1992;216:454-62.
[11] Graziano JN. Cardiac anomalies in patients with congenital diaphragmatic hernia and their prognosis: a report from the Congenital Diaphragmatic Hernia Study Group. J Pediatr Surg 2005;40:1045-50.
[12] Neville HL, Jaksic T, Wilson JM, et al. Fryns syndrome in children with congenital diaphragmatic hernia. J Pediatr Surg 2002;37(12):1685-7.
[13] Lally KP, Paranka MS, Roden J, et al. Congenital diaphragmatic hernia: stabilization and repair on ECMO. Ann Surg 1992;216:570-3.
[14] Downard CD, Betit P, Chang RW, et al. Impact of Amicar on hemorrhagic complications of ECMO: a ten-year review. J Pediatr Surg 2003;8(8):1212-6.
[15] Horwitz JR, Cofer BR, Warner BW, et al. A multicenter trial of 6-aminocaproic acid (Amicar) in the prevention of bleeding in infants on ECMO. J Pediatr Surg 1998;33:1610-3.
[16] The Congenital Diaphragmatic Hernia Study Group. Estimating disease severity of congenital diaphragmatic hernia in the first 5 minutes of life. J Pediatr Surg 2001;36(1):141-5.
[17] Heiss KF, Clark RH. Prediction of mortality in neonates with congenital diaphragmatic hernia treated with extracorporeal membrane oxygenation. Crit Care Med 1995;23(11):1915-9.